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Urbanization as a facilitator of gene flow in a human health pest

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Abstract

Urban fragmentation can reduce gene flow that isolates populations, reduces genetic diversity and increases population differentiation, all of which have negative conservation implications. Alternatively, gene flow may actually be increased among urban areas consistent with an urban facilitation model. In fact, urban adapter pests are able to thrive in the urban environment and may be experiencing human-mediated transport. Here, we used social network theory with a population genetic approach to investigate the impact of urbanization on genetic connectivity in the Western black widow spider, as an urban pest model of human health concern. We collected genomewide single nucleotide polymorphism variation from mitochondrial and nuclear double-digest RAD (ddRAD) sequence data sets from 210 individuals sampled from 11 urban and 10 nonurban locales across its distribution of the Western United States. From urban and nonurban contrasts of population, phylogenetic, and network analyses, urban locales have higher within-population genetic diversity, lower between-population genetic differentiation and higher estimates of genetic connectivity. Social network analyses show that urban locales not only have more connections, but can act as hubs that drive connectivity among nonurban locales, which show signatures of historical isolation. These results are consistent with an urban facilitation model of gene flow and demonstrate the importance of sampling multiple cities and markers to identify the role that urbanization has had on larger spatial scales. As the urban landscape continues to grow, this approach will help determine what factors influence the spread and adaptation of pests, like the venomous black widow spider, in building policies for human and biodiversity health.

KEYWORDS
pest, popgraph, social network, spider, urban facilitation

1 | INTRODUCTION

By 2050, two-thirds of the human population are predicted to live in urban areas (United Nations, 2015). In the United States, the most rapid urban growth occurs in the west, where urban centres expand outward into pristine natural habitat (Seto, Fragkias, Gueneralp, & Reilly, 2011; Seto, Gueneralp, & Hutyra, 2012; US Census, 2010). The loss of these natural habitats reduces local and global biodiversity, which has implications for conservation and human health (Keyghobadi, 2007; McDonald, Kareiva, & Formana, 2008; McKinney, 2002, 2006). Although negative eco-evolutionary consequences of urbanization have been documented (Faeth, Bang, & Saari, 2011; Johnson & Munshi-South, 2017; McDonald et al., 2008; McDonnell & Hahs, 2015; McKinney, 2006; Shochat et al., 2010), we know very little about species that thrive in urban ecosystems, known as urban adapters (Blair, 1996; Shochat, 2004). Urban adapters thrive in urban areas because they are able to take advantage of new or more abundant resources, and also have increased opportunities for higher

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population connectivity, or gene flow on the landscape, due to human-mediated transport (Crispo, Moore, Lee-Yaw, Gray, & Haller, 2011). Although urban adapters can have demonstrated benefits for humans, many are pests that cause structural damage, mental anguish and human health concerns (Booth et al., 2012; Crissman et al., 2010; Puckett et al., 2016). As urban areas continue to expand, an understanding of what factors influence how urban pests disperse and adapt to the urban landscape is necessary for the control and management of pest species (Booth et al., 2012; Crissman et al., 2010; Hauser & McCarthy, 2009; Menke et al., 2010; Puckett et al., 2016; Saenz, Booth, Schal, & Vargo, 2012).

From an eco-evolutionary perspective, the urban fragmentation model of gene flow predicts populations become isolated because of urbanization fragmenting the landscape (Allendorf & Luikart, 2007; Debinski & Holt, 2000; Holderegger & Di Giulio, 2010; Keyghobadi, 2007; Storfer, Murphy, Spear, Holderegger, & Waits, 2010; Trizio et al., 2005; Vandergast, Bohonak, Weissman, & Fisher, 2007; Vandergast et al., 2009). These isolated populations have reduced dispersal between patches that lead to reduced gene flow, which is expected to lead to increased drift, reducing genetic diversity within patches and increasing genetic differentiation among them (Keyghobadi, 2007). For example, white-footed mice (*Peromyscus leucopus*) in urban parks experience high genetic differentiation between parks, and as canopy cover decreases with increasing impervious surfaces, genetic diversity is reduced (Munshi-South, 2012; Munshi-South & Karchenko, 2010). This trend is consistent across a variety of taxa, including both vertebrates and invertebrates (Beninde et al., 2016; Booth, Montgomery, & Prodőhl, 2007; Davis, Murray, Fitzpatrick, Brown, & Paxton, 2010; Lourenco, Alvarez, Wang, & Velo-Anton, 2016). However, as urban adapters may take advantage of these new urban habitats as corridors to dispersal, studies have invoked the alternative urban facilitation model that predicts increased dispersal within and between urban areas (Crispo et al., 2011). Under this model, not only would higher gene flow among urban areas be expected compared to that among nonurban areas (Crispo et al., 2011), but evidence of isolation by distance (IBD) is expected to be weak as urban-mediated human transportation can result in adapter populations that are distantly geographically separated being more genetically similar. For example, human commensals such as bed bugs and German cockroaches show low population structure across global geographic areas due to their high association with human expansion and colonization (Booth, Balvín, Vargo, Vilímová, & Schal, 2015; Booth et al., 2012; Vargo et al., 2014). Thus, while urban fragmentation studies have dominated the literature, our understanding of how urbanization facilitates gene flow and increases genetic diversity is particularly poor despite the utility this model has for urban adapter pests and applications to conservation and human health.

In characterizing gene flow patterns on variable spatial and temporal scales, interdisciplinary approaches in landscape, population and evolutionary genetics have emerged (Dyer, 2015). For example, traditional pairwise measures such as F_{ST} are typically used to impart information on genetic differentiation between sampled populations. However, measures such as conditional genetic distance (cGD), which is derived from population networks that are generated under principles of network theory, inform about connections among all sampled populations, and in fact, outperform other genetic distance metrics (Dyer & Nason, 2004; Dyer, Nason, & Garrick, 2010). Social network theory has gained significant exposure in disciplines such as sociology (Easley & Kleinberg, 2010), economics (Seiler, Collins, & Fefferman, 2014), ecology (Greenbaum, Hoffman, Shalev, & Zelnik, 2015) and evolutionary biology (Greening & Fefferman, 2014; Pickrell & Pritchard, 2012); however, this approach has been unexplored in the context of the human impact of urbanization, despite urban areas being models of social networks that reflect human interactions. Instead of simply identifying evidence of population structure overall, these analyses would be invaluable to our understanding of how urban areas act as a biological network with specific connections identified that both fragment and facilitate gene flow among urban pest populations.

The Western black widow spider, *Latrodectus hesperus*, is a perfect urban eco-evolutionary model because it (a) inhabits an area of rapid Western U.S. urbanization, (b) is recognized as an urban adapter due to its recent expansion and success in urban areas from its native desert habitat, (c) maintains a large geographic distribution among multiple urban and nonurban areas (Figure 1) and (d) is an urban pest with significant medical relevance, which has implications for the evolving social and physical interactions between humans and our natural environment. We have previously documented urban ecological differences in fertility, behaviour, web-building and diet (Johnson, Trubl, & Miles, 2012; Trubl, Gburek, Miles, & Johnson, 2012), as well as more dense urban aggregations, which are of

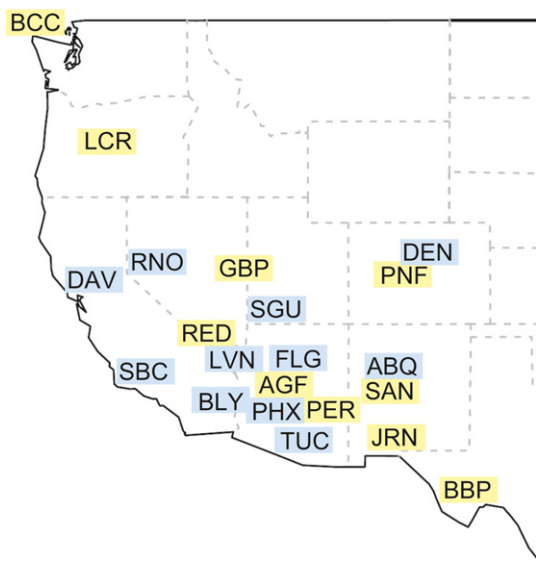

FIGURE 1 Western black widow spider geographic distribution. Distribution of the sampling locales, available from the geographic range of the Western black widow spider across the Western United States. Abbreviations highlighted in blue and yellow reflect urban and nonurban locale sampling, respectively (Supporting Information Table S1)

health concern given its highly toxic venom (Vetter & Isbister, 2008). There is sex-biased dispersal in this species where adult female widow spiders are the sedentary sex, build a web as a juvenile and only migrate when resources are depleted, whereas males and spiderlings have the potential for aerial "ballooning" dispersal (Chamberlin & Ivie, 1935). Although there are well-documented ecological differences between urban and nonurban Western black widow spiders, we know little about the evolutionary potential of these ecological differences and how they may impact humans, especially as secondary contact among these habitats continues to be inevitable. In this respect, evolutionary population genetic analyses are needed to address how urbanization fragments or facilitates gene flow for this pest species.

Here, we used population genetic and social network analyses to investigate the impact of anthropogenic disturbance on connectivity in a model urban pest with human health concerns. Specifically, we conduct population structure, phylogeographic, genetic connectivity and network analyses of the Western black widow spider across multiple urban and nonurban locales using both mitochondrial and genomewide nuclear ddRAD-seq markers. In addition to the null model of urban fragmentation that is typically invoked, we also consider whether urban areas have acted to facilitate gene flow (e.g., via human-mediated transport). If the urban facilitation model drives evolution for urban adapter pests like the Western black widow spider, then we hypothesize that urban locales have higher genetic diversity, less population structure, more recent admixture, higher phylogeographic similarity, higher genetic connectivity and overall lower IBD in contrast to nonurban locales.

2 | MATERIALS AND METHODS

2.1 | Sampling

The Western black widow spider (*L. hesperus*) is a nocturnal web-building predator that is both asocial and highly cannibalistic in all life stages. In urban areas, populations can be densely aggregated with abundant food resources of crickets and cockroaches in open xeric-landscaping, whereas their nonurban distribution is very patchy, isolated and associated with arid, rocky outcrops and dry river-bed banks that are highly sheltered. In considering these distributions, the difficulty in access to Western black widow spider habitat, and our objective in making contrasts between multiple urban and nonurban locales, we assembled a sample of 210 individuals from 11 urban and 10 nonurban locales (10 individuals from each locale) spanning the Western United States (Figure 1, Supporting Information Table S1). Each locale constituted an area of ~0.5 km sq. As males are significantly smaller and rarely found, our sample is almost completely of females (<10 males). For comparative phylogenetic analyses, we also sampled three Southern black widow spiders (*Latrodectus mactans*) from Richmond, VA to be used as an outgroup as this species is the sister taxon to *L. hesperus* (Garb, Gonzalez, & Gillespie, 2004). After collection, samples were immediately stored in 90% EtOH and then placed at −20°C.

2.2 | DNA sequence collection

Both mitochondrial DNA (mtDNA) and nuclear DNA (nuDNA) were collected to tease apart temporal and spatial characteristics of gene flow from demographic contributions, that is, maternally inherited and more-rapidly evolving mtDNA provides the opportunity to test for patterns of potentially further reduced gene flow independent of urbanization. Using tissue dissected from individuals' legs, DNA was extracted using the DNeasy Blood & Tissue kit (Qiagen). For both *L. hesperus* and *L. mactans*, the mtDNA ND1 region was collected from PCR-amplified fragments of 480 bp using previously published primers (Hedin, 1997). PCR products were purified by treatment with Exonuclease I and Shrimp Alkaline Phosphatase (US Biochemicals) and then sequenced using an Applied Biosystems 3730 capillary sequencer. Sequences were aligned and edited manually using SEQUENCHER v3.1.1 (GeneCodes).

For both *L. hesperus* and *L. mactans*, genomewide nuDNA fragments were generated from double-digest RAD sequencing (ddRAD-seq) according to the protocol outlined in Peterson, Weber, Kay, Fisher, and Hoekstra (2012). Each pool of 20 individuals was sequenced on a single-end, 100-bp Illumina HiSeq 2500 lane. The Illumina reads were processed to identify and genotype loci across all individuals using the STACKS v1.44 pipeline (Catchen, Amores, Hohenlohe, Cresko, & Postlethwait, 2011; Catchen, Hohenlohe, Bassham, Amores, & Cresko, 2013) with program parameters set to default unless otherwise noted. The raw fastq files were demultiplexed, filtered for quality (Phred quality score >10) and the presence of barcodes, trimmed to 90 bp in length, and filtered for reads that did not contain the *Eco*RI recognition site using *process_radtags*. Because a well-annotated genome is not yet available for *L. hesperus* or a closely related species, a de novo assembly of raw reads into RAD tags was generated using *ustacks*, with the minimum number of reads set at $m = 5$. Next, a catalog of consensus loci was generated using *cstacks* with the number of mismatches allowed between tags set to $n \leq 5$. After alleles were identified for each individual against the catalog using *sstacks*, the data were further filtered using *populations* with a minimum coverage of 5× per allele for each individual. One single nucleotide polymorphisms (SNP) per fragment was randomly chosen to reduce effects of locus-specific natural selection and clustering due to LD. Genotype data were exported from STACKS in each of the formats needed for subsequent analyses.

2.3 | Data analyses

Estimates of nucleotide diversity as the average number of pairwise differences (π), the number of polymorphic sites (S) and distributions reflecting contrasts between these two (Tajima's *D*, 1989) were estimated within all locales and for our groupings of urban and nonurban samples. For estimates of genetic differentiation, standard pairwise F_{ST} estimates and tests for statistical significance were calculated between all locales, as well as for comparisons of urban and nonurban samples. Principal component analysis (PCA) was used to make inferences about clustering of individuals, with respect to

urban and nonurban locales, as this analysis is model-free (Jombart, 2008; Novembre & Stephens, 2008; Jombart, Pontier, & Dufour, 2009; Jombart, Devillard, & Balloux, 2010). The PCA was generated in the GSTUDIO package (Dyer et al., 2010) in R.

The program BEAST v2.4.5 (Bouckaert et al., 2014) was used to generate *L. hesperus* haplotype tree topologies, with our *L. mactans* samples as an outgroup, to estimate phylogeographic relationships among urban and nonurban individuals and locales. All analyses were performed using a TrN+G model of substitution, identified here as the appropriate model using JMODELTEST (Posada, 2008), on the basis of the Akaike information criterion. To calibrate the mtDNA tree, BEAST v2.4.5 was first run using published CO1 mtDNA sequence data and mutation rate estimates (Garb et al., 2004), and the fossil calibration divergence of the *Latrodectus* clade estimated at ~65 MYA (as in Dimitrov et al., 2012). Using a similar iterative approach employed by others using fossil calibrations (Boykin, Bell, Evans, Small, & De Barro, 2013; Dornburg, Brandley, McGowen, & Near, 2012; Heled & Drummond, 2012; Qu et al., 2014; Valente et al., 2012), our initial BEAST run used the relaxed uncorrelated lognormal clock estimate and speciation yule process model, with five chains for 5,000,000 generations (with the first 5,000 discarded as burn-in) and logging every subsequent 5,000. Once the estimate for divergence between the *L. hesperus* and *L. mactans* clade was generated, and because mtDNA regions are inherently linked, we set the T_{MRCA} at the node estimated from the published CO1 data for our ND1 sequence data here. This run used the lognormal relaxed clock (uncorrelated) estimate and coalescent constant size model, with five chains for 50,000,000 generations (with the first 5,000 discarded as burn-in) and logging every subsequent 5,000. All runs were checked for convergence of the chains in the program TRACER 1.5 (Drummond & Rambaut, 2007). Log files for each run were combined using LOGCOMBINER, and a consensus tree was summarized using TREEANNOTATOR v1.6.1 (Drummond & Rambaut, 2007). For our nuDNA tree, we again used BEAST with our *L. mactans* samples as an outgroup. We used the same run parameters as above (except for mutation rate, Masta, 2000) and visualized both the mtDNA and nuDNA data sets using FIGTREE (Rambaut, 2012).

To further place our phylogenetic results in a demographic perspective (e.g., temporal changes in population size), the program MS (Hudson, 2002) was used to generate simulations under a coalescent model (i.e., given sample sizes and estimates of θ in overall sample). We simulated multiple models of migration between and among urban and nonurban locales, by varying migration rates and population sample sizes. Within these simulations, we also varied the coalescent time for urban populations. Since black widow spiders have approximately one generation per year (Herms, Bailey, & McInor, 1935), and urbanization in the Western United States emerged ~200–500 years ago (US Census, 2010), we set the number of generations since divergence between 200 and 500. After each set of simulations, we calculated all pairwise F_{ST} values as well as other summary statistics of nucleotide diversity and determined statistically significant differences by comparing observed and simulated data using an in-house R script.

Genetic connectivity among sampled locales was estimated from the conditional genetic distance statistic cGD (Dyer & Nason, 2004), which is estimated from the genetic covariance among locales. This measure of genetic covariance among all locales can be visualized as a popgraph using the POPGRAPH R package (Dyer et al., 2010), where nodes represent sampled locales and edges represent genetic connections among locales. Popgraph topology is not only a visualization of cGD as genetic covariance but also of social network parameters that define the popgraph. These social network analyses evaluate genetic relationships among locales and relative contributions of key "actors" using mathematical graph theory (Wasserman & Faust, 1994), which here, visually represent gene flow among all sampled locales to identify hubs of higher connectivity on the landscape. Social network node-specific parameters including closeness, degree, betweenness and eigenvector centrality were estimated from the popgraphs using the POPGRAPH R package. "Closeness" measures the degree to which a node is genetically similar to all other nodes in the network, where higher closeness values indicate further genetic distance to the next node. "Degree" is the number of edges a node has connecting it to other nodes. "Betweenness" is the sum of the shortest paths (i.e., the combination of edges among multiple nodes), where higher betweenness values indicate more paths that pass through a node. Eigenvector "centrality" computes the extent to which each node is centrally located among all other nodes within the popgraph topology. To identify statistical differences and congruence between mtDNA and nuDNA popgraph topologies, as well as differences within these topologies with respect to urban and nonurban contrasts, we performed *t* tests for all of these network parameters, with Bonferroni corrections for multiple comparisons.

Finally, to test for signatures of IBD, Euclidean geographic distances were estimated from latitude and longitude coordinates using the FIELDS package (Nychka, Furrer, Paige, & Sain, 2015) in R, and genetic distances were calculated as cGD (see above). Mantel tests were performed on the geographic and genetic distances in R.

3 | RESULTS

We generated >100K SNPs, from which a filtered high-quality data set of 40,533 SNPs from nuDNA and 124 SNPs from mtDNA sequences were analysed. Estimates of nucleotide diversity for both marker types (Supporting Information Table S2) are high and consistent with arthropod studies (Burns et al., 2017; Garb et al., 2004). Genetic diversity for the overall urban sample was lower than for the nonurban sample, and this was true for both mtDNA and nuDNA data sets. However, we find a significant excess of rare alleles for the urban sample in both the mtDNA (Taj $D = -1.44$, $p < 0.01$) and nuDNA (Taj $D = -1.57$, $p < 0.01$) data sets. In fact, when we examined how variation is distributed within locales, we found that although average locale mtDNA diversity was similar for urban and nonurban samples ($t_{16} = 0.83$, $p = 0.42$), the average nuDNA diversity was significantly higher within urban locales compared to within nonurban locales ($t_{14} = -1.66$, $p = 0.02$).

The average F_{ST} of all pairwise comparisons for mtDNA and nuDNA was 0.53 ± 0.02 and 0.23 ± 0.01, respectively, and these two estimates were significantly different (t_{391} = −9.85, p < 0.001; Supporting Information Figure S1). Although F_{ST} estimates are significantly high for both the urban and nonurban data sets, urban locales are significantly less-genetically differentiated from each other for mtDNA variation, in contrast to that observed between nonurban locales (F_{ST} = 0.42 ± 0.04 vs. 0.56 ± 0.05; t_{96} = −1.33, p < 0.01). Although the overall level of differentiation is lower than that seen with mtDNA variation, the nuDNA variation shows the same significant pattern in the contrast of urban vs. nonurban locales (F_{ST} = 0.15 ± 0.03 vs. 0.30 ± 0.04; t_{90} = −0.99, p < 0.05).

The PCA of SNP genotypes for each marker data set identified different patterns of genetic structure among individuals. For mtDNA, urban locales are predominantly clustered (Figure 2a), while nonurban locales are individually more isolated, shown by PC1 and PC2 explaining 55% of the genetic variance (Figure 2b). Further PCs explain significantly less variance with no additional separation of urban or nonurban locales. The nuDNA variation exhibits far less genetic clustering for both urban (Figure 2a) and nonurban (Figure 2b) locales compared to mtDNA variation, with PC1 and PC2 explaining only 27% of the genetic variance. Further inspection of PCs resulted in far less variance explained and even less separation compared to that seen with the mtDNA data set.

Our phylogenetic analysis (Figure 3) of mtDNA haplotype variation shows that an ancestral clade is dominated by monophyletic groups of nonurban locales with deep evolutionary divergence. In contrast, urban locales form a predominant phylogenetic clade of mixed lineages more recently derived from nonurban ones. In fact, our MS simulations under different demographic scenarios indicate that observed urban genetic diversity is significantly more comprised of rare rather than common haplotypes (Taj D = −2.05, p < 0.001), compared to nonurban locales (Taj D = −0.92, ns). On the other hand, the phylogenetic analysis of the nuDNA data set resulted in

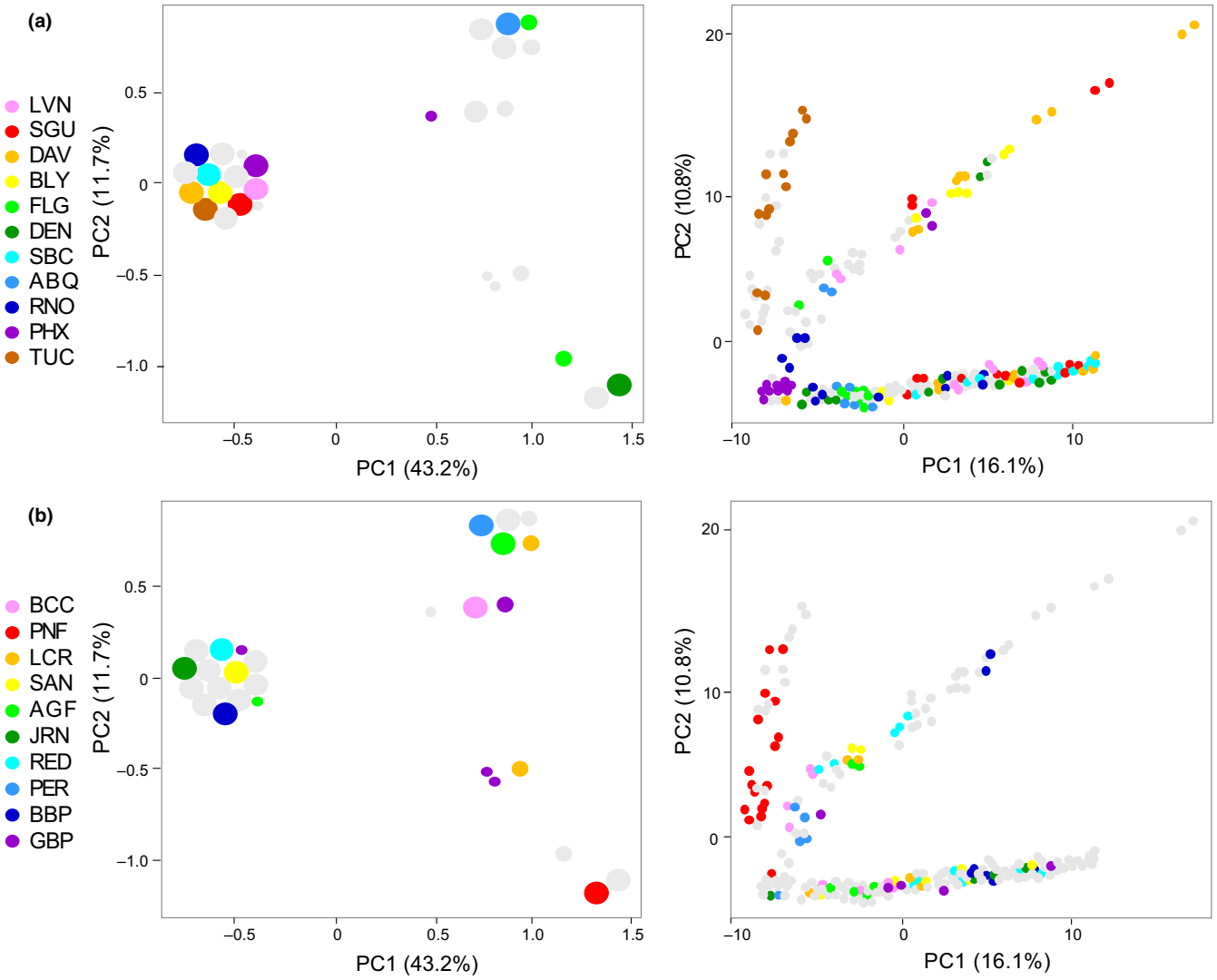

FIGURE 2 Principal component analysis (PCA) biplot. (a) PCA of mitochondrial DNA (mtDNA) (left) and nuclear DNA (nuDNA) (right), with urban samples highlighted, and (b) PCA of mtDNA (left) and nuDNA (right), with nonurban locales highlighted. Each circle size reflects relative number of individual samples at that point (see Supporting Information Table S1 for locale abbreviations)

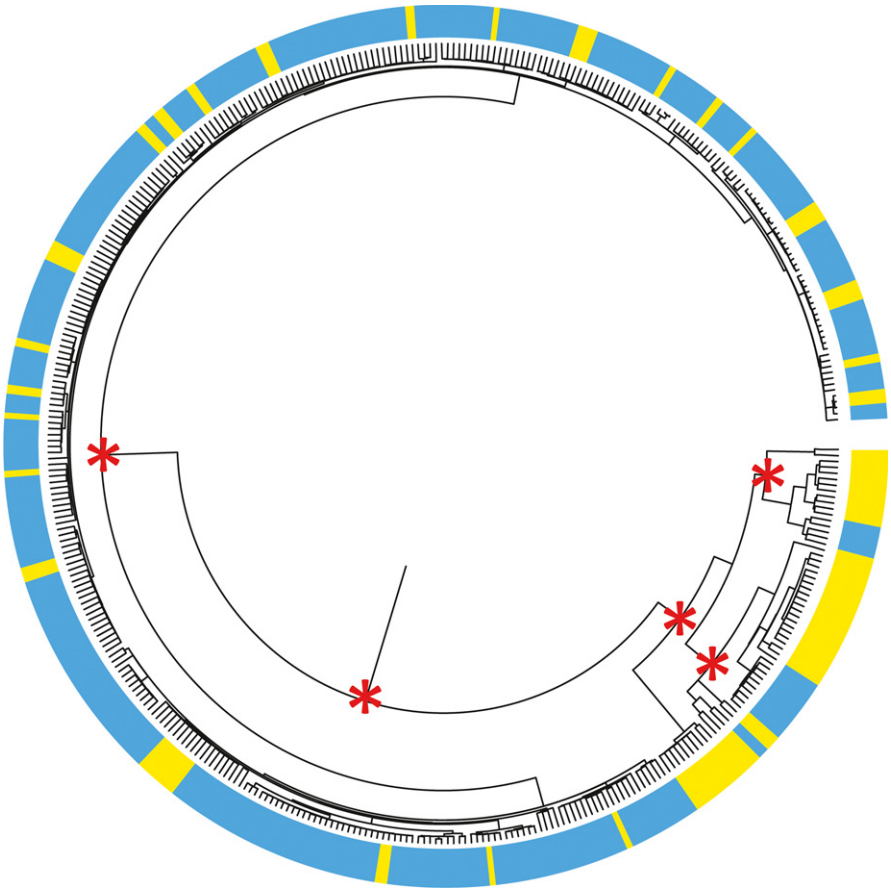

FIGURE 3 Phylogenetic tree. BEAST analysis of 210 mitochondrial DNA (mtDNA) haplotypes with highlights in blue and yellow reflecting urban and nonurban locale sampling, respectively. Major node support as posterior probabilities above 70% is noted as red asterisks

no statistical support for any clade structure, whether they correspond to urban or nonurban locales.

Our popgraph topologies for mtDNA and nuDNA data sets show contrasting patterns of genetic connectivity (Figure 4). Although our cGD estimate was significantly higher in overall mtDNA than nuDNA data set (t_{68} = 9.50, $p < 0.0001$), this estimate among urban locales is higher than among nonurban locales for both marker types (mtDNA t_{55} = 3.30, $p < 0.001$; nuDNA t_{97} = 1.70, $p < 0.05$). The Mantel tests found weak associations for IBD analyses for both the mtDNA (r^2 = 0.030, p = 0.02) and nuDNA (r^2 = 0.008, p = 0.15) data sets. When Mantel tests were applied individually to the urban and nonurban data sets, no significant patterns of IBD emerged.

The social network parameters underlying the popgraph also have contrasting patterns between mtDNA and nuDNA data sets (Supporting Information Table S3). "Closeness" is significantly higher in the mtDNA popgraph for nonurban than urban locales (t_{15} = −2.50, $p < 0.001$) and is also higher (which again, means greater distance) overall in the mtDNA popgraph than the nuDNA popgraph (t_{20} = −18.18, $p < 0.00001$). Overall, there are a fewer number of connections, and even more visible disconnections, in the mtDNA popgraph compared to the nuDNA popgraph. This result is evidenced by the parameter "degree" being significantly higher in the nuDNA than the mtDNA popgraph (t_{20} = −3.50, $p < 0.0001$); however, we note "degree" is not significantly different between urban and nonurban locales for mtDNA or nuDNA popgraphs. We found no significant results for the "betweenness" parameter analyses overall; however, nonurban locales are among the highest ranked values in the mtDNA popgraph, whereas multiple urban locales are among the highest values in the nuDNA popgraph. The results of "degree" and "betweenness" indicate that while urban locales have higher network connections overall, urban locales do not drive genetic connectivity similarly. In fact, our analyses of "centrality" or hub determinism, which is a function of the aforementioned social network parameters combined, identify Phoenix (PHX), Reno (RNO) and Las Vegas (LVN) as major urban hubs of connectivity, whereas urban locales Albuquerque (ABQ), Davis (DAV) and Denver (DEN) show little to no influence on popgraph network structure.

4 | DISCUSSION

With the Western black widow spider as an urban pest model, this study set out to specifically test whether urbanization primarily acts as a barrier to gene flow, or instead, that urbanization facilitates gene flow for this urban adapter. From multiple perspectives of population, phylogenetic and network analyses, the primary observations here are that in comparison with nonurban locales, urban locales have higher within-population genetic diversity, lower between-population genetic differentiation and higher overall estimates of genetic connectivity. We discuss these results in the light of their

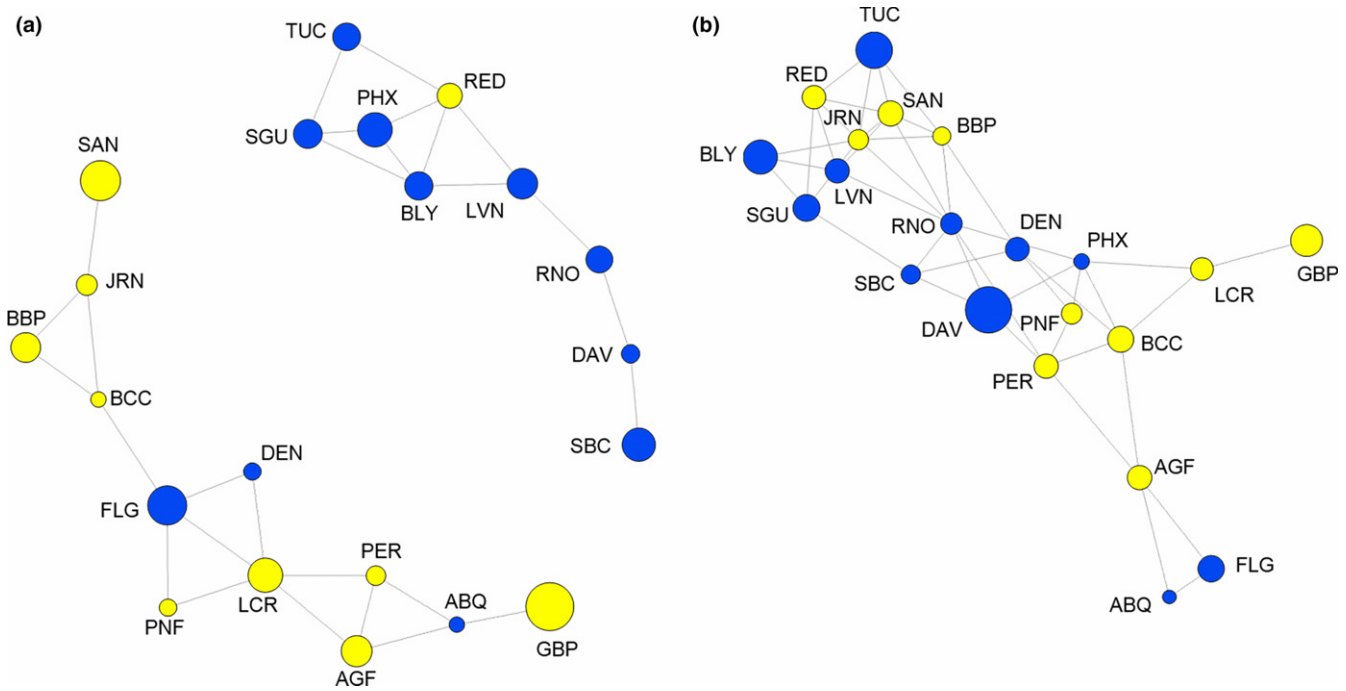

FIGURE 4 Popgraph network analysis. (a) Mitochondrial DNA (mtDNA) popgraph and (b) nuclear DNA (nuDNA popgraph) among urban (yellow) and nonurban (blue) locales (Supporting Information Table S1). The size of each locale node reflects the amount of genetic variance within the locale, and length of connections between nodes is proportional to conditional genetic distance (cGD) as a measure of genetic covariance reflecting gene flow among locales

support for an urban facilitation model, and how these approaches help our perception of conservation and human health in urban environments.

The urban fragmentation model predicts that urbanization acts as a barrier to dispersal (e.g., Holderegger & Di Giulio, 2010; Storfer et al., 2010) and that measures of Western black widow spider genetic differentiation among urban areas would be elevated. Our F_{ST} values at first glance are significantly high for both the mtDNA and nuDNA data sets. In fact, these measures rival that of studies in both plants and animals (e.g., Ascunce et al., 2013; Bartlewicz, Vandepitte, Jacquemyn, & Honnay, 2015; Booth et al., 2015; Hitchings & Beebee, 1997; Johnson & Munshi-South, 2017; Munshi-South, Zak, & Pehek, 2013; Saenz et al., 2012). However, while high dissimilarity indices such as F_{ST} might reflect low gene flow, they do not necessarily reflect a reduction in gene flow due to urbanization. In fact, genetic differentiation among multiple geographically dispersed nonurban locales is significantly greater than that observed among urban locales for both mtDNA and nuDNA data sets. Thus, the contrasts of urban and nonurban patterns actually prove to be more consistent with gene flow being relatively facilitated, and not reduced among urban locales.

Estimates of genetic diversity and phylogeographic history of Western black widow spiders also support the urbanization facilitation model. Although the overall nonurban population has greater genetic diversity than the overall urban population for both mtDNA and nuDNA data sets, on average, urban within-locale genetic diversity is significantly higher than that seen within nonurban locales. These results are surprising given that urban population genetic studies more often find reduced genetic diversity within urban locales. As gene flow reduces, classic theoretical and empirical studies in population genetics predict that while individual nonurban locales may suffer from reduced effective population sizes, increased inbreeding and higher probability of fixation of alleles by drift alone, this process happens randomly across locales (e.g., Dobzhansky, 1937; Wright, 1982). This process results in possibly even higher levels of allelic diversity maintained in the overall nonurban population. On the other hand, the urban population maintains higher within-locale diversity expected from relatively higher gene flow but a lower overall level of genetic diversity as expected if it was more recently derived from nonurban areas. Our phylogenetic analysis and demographic simulations support this exact hypothesis. While the nuDNA data set provides little phylogenetic information owing to the likely higher level of admixture and diversity of biparental variation (more below), the maternally inherited mtDNA preserves the phylogenetic history showing older and ancestral clades of seemingly more evolutionarily isolated nonurban locales. This pattern is in contrast to the urban locales that form a larger, more homogenous and recently derived clade. Thus, while we expected a molecular signature of colonization of urban areas from nonurban areas, what is more interesting is that our demographic simulations also identify a signature of a recent population expansion (i.e., a significant excess of rare alleles) associated with the movement of Western black widow spiders into urban locales. These observations demonstrate not only that these organisms are successfully invading urban

environments, but they do so in rapidly spreading across large geographic areas. These patterns should be noted as a molecular signature of an urban adapter and pest (Booth et al., 2012, 2015; Vargo et al., 2014).

Our final set of analyses that examine population clustering and network characteristics shows the most demonstrable evidence of urban facilitation with higher genetic connectivity overall driven by urban locales. The PCA results show very little clustering for nuclear variation, whereas for the mtDNA haplotypes, patterns are again consistent with the historical isolation of nonurban locales compared to the large clustering of urban locales, consistent with our phylogenetic analysis. However, not seen previously are nonrandom patterns of urban and nonurban haplotypes clustering, which hints at how certain locales influence introgression and gene flow more than others. While our mtDNA and nuDNA popgraphs indicate fewer and greater connections among locales, respectively (as expected by their demographic histories, more below), overall, urban locales have significantly more connections and have higher genetic proximity to other locales than nonurban locales do. Furthermore, this degree of high urban connectivity results in several nonurban locales being connected to each other only via gene flow through certain urban locales on the landscape. This surprising pattern has potentially negative evolutionary impacts for the phenotypes within these historically isolated nonurban locales. In fact, our IBD analysis shows a consequence of urban facilitation is that Western black widow spiders, even from nonurban locales, are more genetically similar than expected given their geographically distant separation. Altogether, this popgraph evidence hints at certain urban areas as drivers of this increased genetic connectivity through both urban and nonurban locales.

A final step is to test hypotheses of whether specific locales and potential factors influence urban genetic connectivity. Our network parameter analyses identified Phoenix, Reno and Las Vegas as major urban hubs, whereas other urban locales such as Denver have little influence on genetic connectivity. Thus, while we find urbanization overall may facilitate gene flow, not all urban locales behave the same in this role. Multiple urban features have been implicated as factors driving urban gene flow such as percent impervious surface, canopy cover or human population size (Alberti, 2015; Johnson & Munshi-South, 2017; McDonnell & Hahs, 2015). However, these studies are often focused on fine-scale gene flow within a single urban area (Emaresi, Pellet, Dubey, Hirzel, & Fumagalli, 2011; Munshi-South, 2012; Van Buskirk, 2012; Sacks, Brazeal, & Lewis, 2016), whereas our study focused on characterizing broadscale genetic connectivity among multiple urban locales. With this in mind, we expect human population size to be a mitigating factor, as human-mediated transport among these areas may be tied to the simple volume of humans in each urban area. That said, while Phoenix and Denver have high population sizes, they have contrasting high and low influence on genetic connectivity, respectively; yet, Reno and ABQ have low population sizes, but have high and low influence on genetic connectivity, respectively (U.S. Census, 2010; Supporting Information Table S3). Interestingly, these four urban areas also differ with respect to age of human colonization and population expansion, as well as their proximity along urban corridors. These social network analyses help identify not only drivers of gene flow, but help test hypotheses of how landscape features interact in complex ways.

The overall population genetic pattern shown here is consistent with an urban facilitation model; nonetheless, we still observe clear signatures of sex-biased dispersal patterns. Compared to nuDNA, the mtDNA data set for nonurban locales reveals patterns of reduced gene flow, including lower within-locale genetic diversity, higher between-locale differentiation and higher locale clustering. The phylogenetic analyses show strong evidence of population and temporal structure consistent with historical nonurban isolation and more derived urban expansion, but only for the mtDNA data set, which validates life history observations of females being the sedentary sex (Chamberlin & Ivie, 1935). One possible problem in urban landscape studies is that sex-specific spatial autocorrelation, when not accounted for, may result in erroneously concluding that urbanization reduces dispersal distances (as shown by Brashear, Ammerman, & Dowler, 2015). In fact, high site fidelity is common for the majority of web-building spiders (Foelix, 2014), so without different genetic markers to contrast, it is possible that sex-biased dispersal does dampen signatures of urban facilitation overall. However, as our sample is almost completely derived from females, our comparative data sets of mtDNA- and nuDNA-specific patterns enable us to identify strong signatures of urban facilitation despite high site fidelity for females. In addition, our network analyses found that higher gene flow among urban areas has had significantly less influence on mtDNA than nuDNA variation. Thus, although sex-biased dispersal may be looked at as a caveat in gene flow studies, comparative marker and network approaches can reveal the impact that sex-dispersal and urban facilitation each independently have on population variation.

The impacts that urban facilitation has on the evolutionary ecology of urban adapters can be addressed given our choice of the Western black widow spider. The majority of spider species are able to aerially disperse through ballooning as spiderlings (Bell, Bohan, Shaw, & Weyman, 2005), and this dispersal mechanism that may enable them to successfully colonize new areas makes spiders ideal models for studying gene flow into novel urban habitats. However, there is a paucity of studies that have investigated urbanization impacts on arachnids. For one species of wolf spider, urbanization appears to act as a barrier to gene flow (Reed, Teoh, Stratton, & Hataway, 2011), while for another, urbanization appears to facilitate gene flow (Colgan et al., 2002). Additionally, another has shown that an endangered species of tarantula experiences human-mediated gene flow (Machkour-M'Rabet, Henaut, Calme, & Legal, 2012). Given the implications urbanization has for conservation priorities, our work supports the need for population genetic studies that target urban adapters across spatial scales to document the impact of continued urban growth.

It is unknown to what extent the varying degrees of population divergence identified here among nonurban locales has led to local adaptation (e.g., Wright, 1982), and how the now recent secondary

contact among these locales, ushered in by urban gene flow across broad geographic distances, has altered these phenotypes. In fact, if selective pressures differ sufficiently among urban and nonurban, and even among nonurban environments, "urban ecotypes" may evolve and easily sweep across geographically distant urban areas as a result of human-mediated transport (Krtinic, Ludoski, & Milankov, 2012; Schapira & Boutsika, 2012). As one example previously noted, our group has documented Western black widow spider behavioural differences between urban and nonurban habitats, where urban spiders are more densely aggregated, larger, and are more aggressive (Johnson, Gburek, & Stevens, 2017; Johnson et al., 2012; Trubl et al., 2012). Our research here shows that these behavioural changes together with the potentially lethal venom these spiders wield present one such potential urban ecotype. These ecotypes may be the result of coadapted gene and phenotype complexes that are emerging in urban environments due to increased gene flow among urban areas.

5 | CONCLUSIONS

Our study raises several larger questions with respect to both population genetics and functional evolutionary ecology in urban settings. One thought is with respect to how cities are true "replicates" of each other. While they share some characteristics, they can vary in how they impact gene flow, including effects of human population size. Differences in urbanization effects among different cities have been documented previously (e.g., Thompson, Renaudin, & Johnson, 2016); however, without contrasts of multiple urban and nonurban areas, we may erroneously conclude patterns are consistent with an urban fragmentation model. This point begs the question of how common urban facilitation may be, especially with respect to urban adapter pests. Currently, urban population genetic studies tend to focus on fine-scale within city boundaries or on broad scales between cities, whereas studies combining genomewide sampling at both scales are necessary. While previous population genetic studies have studied pest invasion, we find that network analyses can identify specific hubs that can influence pests' spread on the natural landscape. Together with future phenotypic studies across multiple urban environments, this network approach would enable us to identify those traits most associated with urban adaptation, as we seek to reduce spread of pests with negative medical impacts, or on the other hand, increase dispersal of organisms currently under conservation management.

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DATA ACCESSIBILITY

All data and analyses have been deposited in Dryad Digital Repository https://doi.org/10.5061/dryad.dg32560 and are available from the corresponding author upon reasonable request.

AUTHOR CONTRIBUTIONS

The study was designed by all authors. DNA sequencing was performed by L.S.M. The analyses were performed and the manuscript was written by L.S.M. and B.C.V. All authors contributed to the revision of the manuscript.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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